

Kathmandu University
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Department of Artificial Intelligence

Final Year Project Presentation

MarketGyan

NEPSE and Financial News Analysis
An Extension of Khabar AI

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Outline

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2. Objectives and scope
3. Literature review
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Project idea

MarketGyan extends Khabar AI into a Nepal-focused financial NLP assistant for NEPSE-related news, evidence search, and grounded market reports.

- ▶ Connects market snapshots, financial news, and policy documents.
- ▶ Classifies whether a story is directly relevant, indirectly relevant, or not relevant to NEPSE.
- ▶ Produces structured impact fields: event type, direction, sector, symbol, mechanism, and evidence sentence IDs.
- ▶ Uses RAG so current facts remain in retrieved evidence rather than model weights.

Motivation and Background

Why this matters

- ▶ NEPSE data, financial news, and regulatory updates are scattered across many sources.
- ▶ Investors must manually connect market movement with related events.
- ▶ Generic news summarizers do not know NEPSE-specific symbols, sectors, and event taxonomy.

Need for the system

A Nepal-specific assistant should answer with source-backed evidence, not generic positive or negative tone.

Safety boundary

Informational analysis only. No buy/sell advice.

Project Objectives

Main objective

Build and evaluate a Nepal-specific financial news analysis system that combines structured NEPSE data, financial-news retrieval, and a fine-tuned instruction model.

Specific objectives

1. Collect and manage NEPSE-related market data, financial news, and selected regulatory sources.
2. Create a bilingual, evidence-grounded NEPSE impact benchmark.
3. Fine-tune and evaluate Qwen3.5-9B LoRA for compact structured impact extraction.
4. Build a RAG runtime for grounded query answering and report generation.

Scope of the Project

Included

- ▶ Public NEPSE snapshots and finance news.
- ▶ English and Nepali article processing.
- ▶ NEPSE-Impact-500 dataset.
- ▶ Qwen3.5-9B LoRA adapter.
- ▶ Evidence search, Ask MarketGyan, and report generation.

Not included

- ▶ Brokerage integration or order execution.
- ▶ Personalized investment advice.
- ▶ Trading prediction or causal market forecasting.
- ▶ Production deployment until retrieval and content gates pass.

FinBERT

FinBERT shows that adapting transformer models to financial text improves financial sentiment classification compared with generic language models.

- ▶ Useful lesson: financial words are event- and domain-dependent.
- ▶ MarketGyan gap: NEPSE analysis needs more than positive/negative sentiment.
- ▶ Required fields include relevance, event type, impact scope, direction, sector, symbol, and evidence.

Literature Review: Financial LLMs

Study	Relevance to MarketGyan
FinGPT	Supports a data-centric approach: curate financial data and adapt a moderate-size model with lightweight tuning.
FinLLM survey	Highlights hallucination risk, benchmark weakness, and the need to separate validity, grounding, and task metrics.
FinTral	Demonstrates financial instruction tuning for multi-task structured outputs on a moderately sized open model.

Adopted design

MarketGyan treats the Nepal-specific ontology and evidence-grounded benchmark as the main contribution; LoRA is the adaptation method, not the novelty.

Literature Review: Nepali NLP

Work	Relevance and limitation
NepBERTa Subedi et al.	Shows that Nepali-specific pre-training improves Nepali NLP tasks. Evaluates LLMs for Nepali NER and POS tagging, showing promise but also low-resource benchmark gaps.
Thapa et al.	Expands Nepali transformer resources with larger monolingual pre-training for understanding and generation.

Gap

These works are not finance-specific. MarketGyan therefore builds a bilingual NEPSE benchmark instead of relying only on general Nepali NLP resources.

Proposed System Overview

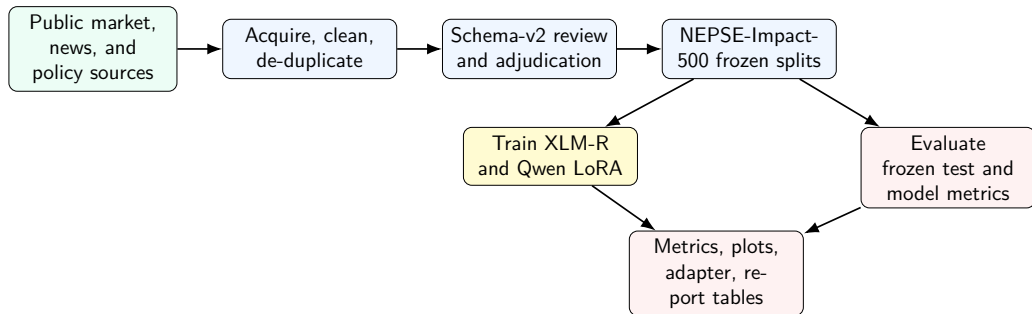
Core pipeline

1. Collect market/news/policy data.
2. Normalize and index finance evidence.
3. Retrieve candidate evidence for a query or report date.
4. Ask Qwen3.5-9B for compact structured output.
5. Materialize citations from retrieved rows.

Final runtime decision

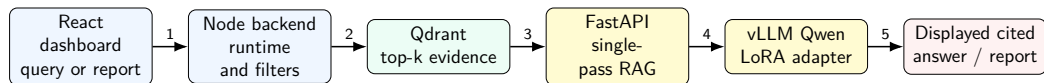
Single-pass RAG replaced the earlier context-heavy multi-agent prompt because it passed live citation checks without exceeding the vLLM context limit.

Training and Evaluation Workflow



The frozen benchmark keeps model comparison separate from live retrieval and frontend behavior.

Daily Inference Workflow



Citations are reconstructed from retrieved evidence rows with URL, excerpt, content hash, chunk ID, and sentence IDs.

Methodology I: Data and Schema-v2

Data pipeline

- ▶ Acquire public NEPSE snapshots, financial news, and selected machine-readable policy text.
- ▶ Clean, de-duplicate, sentence-split, and attach stable URLs/content hashes.
- ▶ Review generated candidates and export only adjudicated gold labels.

Benchmark

- ▶ NEPSE-Impact-500: 300 direct, 100 indirect, 100 hard negatives.
- ▶ Frozen split: 350 train, 75 validation, 75 test.
- ▶ Duplicate groups stay inside the same split to prevent leakage.

Methodology II: Schema-v2 JSON Contract

```
{  
  "relevance": "direct | indirect | not_relevant",  
  "eventType": "market_trading | earnings | regulation | governance |  
               credit_financing | capital_action | sector_industry |  
               project_operations | macro_policy | other | none",  
  "impactScope": "market | sector | company | not_applicable",  
  "impactDirection": "bullish | bearish | neutral | uncertain |  
                     not_applicable",  
  "sectors": ["Banking"],  
  "symbols": ["NABIL"],  
  "rationale": "short evidence-based reason",  
  "confidenceBand": "low | medium | high",  
  "evidenceSentenceIds": ["S1", "S3"]  
}
```

Validation rule

Evidence IDs must point to real numbered source sentences. Relevant rows need defensible impact fields; not-relevant rows must not invent sectors, symbols, or market impact.

Methodology III: Modeling and Runtime Evaluation

Modeling

- ▶ XLM-R relevance and direction baselines.
- ▶ English-only FinBERT sanity baseline.
- ▶ Base Qwen zero-shot and three-shot ablations.
- ▶ Qwen3.5-9B BF16 LoRA targeted-v2 final adapter.

Evaluation

- ▶ Frozen test split for model metrics.
- ▶ Strict structured validity and evidence grounding gates.
- ▶ Per-class F1 for relevance and merged direction.
- ▶ vLLM JSON Schema constrained inference check.

System Requirements and Training Setup

Area	Final setup from report
Local runtime	React, Node/Express, MongoDB, Qdrant, Python FastAPI agent.
Model serving	Lightning AI Studio with vLLM OpenAI-compatible endpoint.
XLm-R baseline	xlm-roberta-base, 512 tokens, LR $2e-5$, max 8 epochs, class-weighted loss, validation macro-F1 selection.
Qwen training	Qwen/Qwen3.5-9B, BF16 LoRA, rank 32, alpha 64, max sequence 2048, LR $2e-5$, 3 epochs, A100 80GB.
Safety	Local report generation, citation validation, disclaimer, no buy/sell advice.

Qwen3.5-9B Training Configuration

Item	Value	Item	Value
Base model	Qwen/Qwen3.5-9B	Precision	BF16 LoRA, no 4-bit
Max sequence	2048	Max generation	256 tokens
LoRA	rank 32, alpha 64, dropout 0.05	Epochs	3
Batch	train/eval 4	Grad accumulation	2
Learning rate	$2e-5$	Optimizer	paged_adamw_8bit
Checkpointing	eval/save every 10 steps	Best model	validation loss
Oversampling	targeted_v2	Train rows	350 to 922 weighted examples
Output	marketgyan-qwen35-9b-a100-80gb -bf16-lora-targeted-v2		

Reason for this setup

The latest run optimized content metrics while preserving strict JSON validity and evidence grounding; the frozen test split was not used for oversampling.

Tools and Technologies

Layer	Tools used
Frontend	React dashboard with MarketGyan tabs: Overview, Ask, Evidence Search, Reports, System.
Backend	Node.js / Express APIs, MongoDB, scheduler, report persistence, review tools.
Retrieval	Qdrant vector database with finance-specific evidence payloads.
ML service	Python, FastAPI, evaluation CLI, schema validators.
Model training	Qwen3.5-9B, LoRA, BF16 training, A100 80GB, frozen test split.
Model serving	Lightning AI Studio, vLLM OpenAI-compatible endpoint, JSON Schema constrained decoding.

Implementation Details

Backend and data modules

- ▶ Article, market snapshot, market document, security, and report records.
- ▶ Candidate generation, review, adjudication, and schema validation.
- ▶ Runtime status, protected report generation, scheduler, and persistence.
- ▶ React tabs for Overview, Ask, Evidence Search, Reports, System, and validation.

ML and runtime modules

- ▶ XLM-R baseline training and Qwen LoRA notebook/script pipeline.
- ▶ Qdrant finance-evidence indexer with source URL, content hash, chunk ID, and sentence IDs.
- ▶ FastAPI single-pass RAG agent for grounded answers and reports.
- ▶ Evaluation CLI for frozen model metrics, retrieval labels, and live scenarios.

Results: 4-bit QLoRA vs 16-bit BF16 LoRA

Metric	4-bit QLoRA (Qwen3-8B)	16-bit BF16 LoRA (Qwen3.5-9B v2)	Status
Structured validity	0.360	1.000	16-bit passes 0.95 gate
Evidence grounding	0.360	1.000	16-bit passes 0.95 gate
Relevance macro-F1	0.384	0.796	Pass
Nepali relevance acc.	0.182	0.788	Large gain
Event macro-F1	—	0.633	Below 0.65
Direction macro-F1	0.413	0.575	Below 0.65
Sector micro-F1	—	0.677	Below target
Symbol micro-F1	—	0.793	Near target
Evidence sentence F1	—	0.736	Below target

4-bit column is the final (v3) Qwen3-8B QLoRA diagnostic run; earlier runs scored 0.413 and 0.347 strict validity.

Why 16-bit BF16 LoRA is the final candidate

QLoRA stayed near 0.35–0.41 strict validity and grounding, below the 0.95 gate. BF16 LoRA reached 1.000 on both and improved Nepali relevance; event, direction, sector, and evidence metrics still need work.

Baseline and Qwen Ablation Results

Supervised baselines

Model / task	Acc.	Macro-F1
XLM-R relevance	0.813	0.743
XLM-R direction	0.567	0.568
FinBERT direction	0.889	0.616

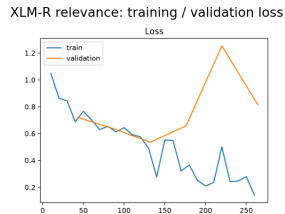
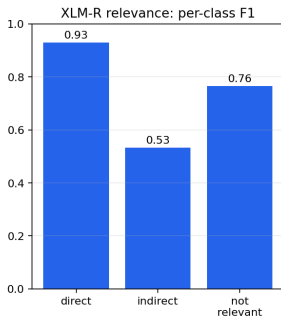
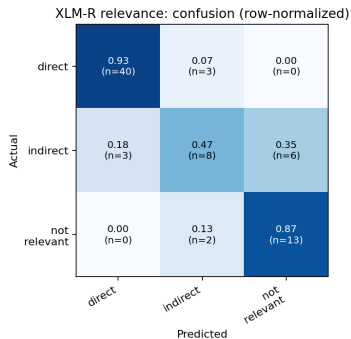
XLM-R direction uses the merged neutral-or-uncertain target; FinBERT support is small and English-only.

Qwen conditions

Condition	Valid	Rel. F1	Evid. F1
Zero-shot	0.000	0.367	0.000
Three-shot	0.987	0.435	0.599
Targeted-v2 LoRA	1.000	0.796	0.736

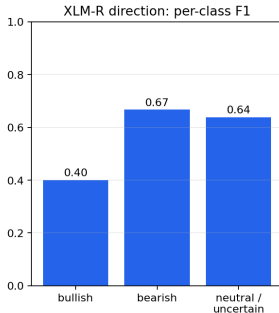
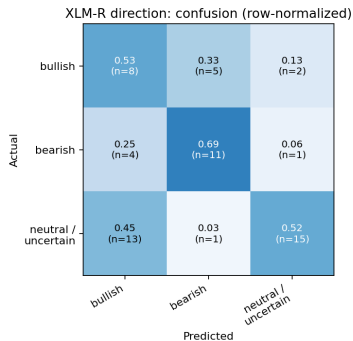
Fine-tuning solves strict structure/grounding and improves relevance, but neutral direction remains weak.

XLM-R Relevance Baseline

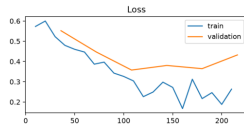


XLM-R relevance: accuracy 0.813, macro-F1 0.743 on the frozen test split. Row-normalized confusion, per-class F1, and the training/validation loss curve.

XLM-R Direction Baseline

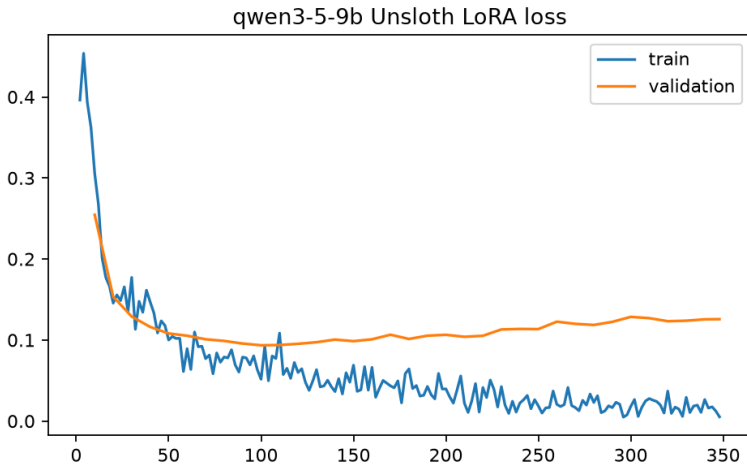


XLM-R direction: training / validation loss



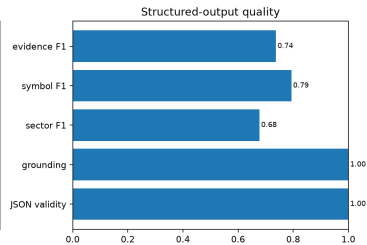
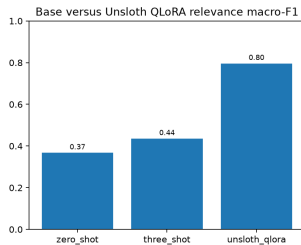
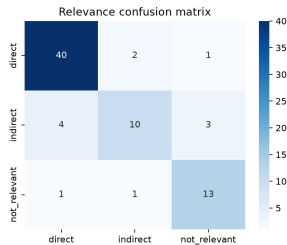
XLM-R direction (merged neutral-or-uncertain): accuracy 0.567, macro-F1 0.568 on relevant frozen-test rows. Merging avoids the unlearnable standalone neutral class and leaves no dead class.

Qwen Targeted-v2 Loss Curve



Validation loss flattens and rises after early improvement, supporting best-checkpoint selection instead of longer blind training.

Qwen Targeted-v2 Test Result Plot



Frozen-test diagnostic plot from the final Qwen3.5-9B targeted-v2 adapter artifact.

Limitations and Future Work

Limitations

- ▶ NEPSE-Impact-500 is small; rare events and neutral direction are under-covered, so standalone neutral cannot yet be learned.
- ▶ Historical market snapshots are partial.
- ▶ Set-valued sector, symbol, and evidence extraction show a recall gap.
- ▶ Lightning-hosted vLLM suits demos, not production deployment.

Future work

- ▶ Expand and adjudicate neutral, sector-industry, regulation, and credit-financing hard cases to move beyond the merged target.
- ▶ Improve sector, symbol, and evidence recall via clearer selection rules and constrained post-processing.
- ▶ Complete independent second annotation and agreement metrics.
- ▶ Keep constrained inference and single-pass RAG while hardening deployment.

Final contribution

MarketGyan delivers a Nepal-specific, evidence-grounded financial NLP research prototype with NEPSE-Impact-500, a Qwen3.5-9B targeted-v2 LoRA adapter, Qdrant evidence retrieval, and a live React/FastAPI/Node demonstration path.

- ▶ The final model passes strict JSON validity and evidence grounding gates.
- ▶ Improved multilingual baselines: XLM-R relevance macro-F1 0.743 and merged-direction macro-F1 0.568 with no dead class.
- ▶ The frontend can demonstrate Ask MarketGyan, Evidence Search, and Reports.
- ▶ Deployment is not recommended until neutral direction and set-valued sector, symbol, and evidence recall improve.

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